A machine-checked Lean 4 / Mathlib formalization of the M9 Newton-edge algebraic core for polynomial continued fractions

(conditional on the analytic edge ⇒ Borel-radius step)

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Abstract

We formalize, in Lean 4 / Mathlib, the *purely algebraic, finitary core* of the M9 Newton-edge lemma for polynomial continued fractions (PCFs) with unit numerator and partial denominator $b(n) = \sum_{k=0}^{d} \beta_k n^k$ ($d \geq 2$, $\beta_d \neq 0$). Four statements are proved with the clean Lean axiom cone {`propext`, `Classical.choice`, `Quot.sound`} and zero `sorry`: (T1) shift-invariance of the leading coefficient, $[\theta^d]\, b(\theta+1) = \beta_d$; (T2) the closed form of the slope-$1/d$ edge polynomial $\chi_d(c) = 1 + (-1)^{d+1}(\beta_d/d^d)c^d$; (T3a) the affine-reindexing coefficient law — the degree-d coefficient of $b \circ (\alpha X + \gamma)$ equals $\alpha^d \beta_d$, which is lead$(b \circ \varphi)$ since deg $= d$ for $\alpha \neq 0$, and is γ-independent; and (T3b) the covariance $\xi_0(\alpha^d \beta_d) = \alpha^{-1}\xi_0(\beta_d)$ of the Borel-singularity radius $\xi_0 = d/\beta_d^{1/d}$ (the T3b covariance is formalized for $\beta_d > 0$; the complex root-modulus identity $|c| = \xi_0$ is *VERIFIED* in computer algebra, not Lean-formalized). The analytic step *Newton edge ⇒ Borel singularity radius* (Wasow §19 sectorial existence; Birkhoff–Trjitzinsky Borel summability) is **out of scope**: it is encoded as an explicit, unproved axiom $\mathsf{H}_{\mathsf{edge}}$ and is *not* discharged. Consequently the Borel-radius end statement is proved *conditional on* $\mathsf{H}_{\mathsf{edge}}$, with that hypothesis visible in its axiom cone by design. This promotes the edge lemma from the source grade "symbolically derived" (a hand induction corroborated by computer algebra) to machine-checked *proven* for its finitary core, while leaving the analytic input honestly labeled. The toolchain is pinned (Lean `v4.29.0`; Mathlib rev `8a17838...`); all axiom cones are reproducible by `#print axioms`. This note is a companion to the M9 deposit (concept DOI `10.5281/zenodo.20542161`).

Epistemic-status legend (used throughout). **PROVEN** = the Lean `#print axioms` cone is contained in {`propext`, `Classical.choice`, `Quot.sound`} *and* contains no `sorryAx`, with zero `sorry`/`admit`/`native_decide` in source. **PROVEN CONDITIONAL on $\mathsf{H}_{\mathsf{edge}}$** = otherwise clean, but the cone additionally contains the explicit axioms `edge_implies_borel_radius` and `borelRadius`. **VERIFIED** = checked numerically/symbolically (computer algebra) but not in Lean. **CITED-CONDITIONAL** = relied upon from the literature and not discharged here. These four grades are kept distinct and the source comments match the realized axiom cones.

1 Scope and the conditional-core discipline

For a PCF with unit numerator and partial denominator $b(n) = \sum_{k=0}^{d} \beta_k n^k$, the convergent denominators Q_n obey the Wallis recurrence $Q_n = b(n)Q_{n-1} + Q_{n-2}$. The generating function $F(z) = \sum_n Q_n z^n$ is annihilated by

$$P \;=\; 1 \;-\; z\,b(\theta+1) \;-\; z^2, \qquad \theta = z\,\frac{d}{dz} \quad (\theta\, z^n = n\, z^n). \tag{1}$$

At $z = 0$ the Newton polygon of P has a single dominant edge of slope $1/d$; the WKB/edge balance on that edge produces the edge polynomial χ_d, whose root modulus is the Borel-singularity radius ξ_0. The passage *edge* $\Rightarrow$ *analytic Borel radius* is classical analysis (Wasow §19; Birkhoff–Trjitzinsky).

This note deliberately separates two things that are easy to conflate:

- the *finitary algebra* underneath the edge — shift-invariance, the closed form of χ_d, and the affine leading-coefficient law — which is purely a statement about polynomials and real powers, and is formalized here as PROVEN;
- the *analytic* edge $\Rightarrow$ radius implication, which is *not* formalized and is taken as the explicit hypothesis $\mathsf{H}_{\mathsf{edge}}$.

This is the "conditional-core" pattern: formalize the finitary core to a clean axiom cone, and take the hard analytic fact as a clearly-labeled hypothesis. The end statement is then honestly *"ξ_0 covariance, conditional on $\mathsf{H}_{\mathsf{edge}}$"*, never silently "$\xi_0$ verified".

2 The four proven statements

All four are in `PcfEdge/Basic.lean`. We give the exact Lean statements (verbatim) followed by the mathematical content.

T1 — shift-invariance of the leading coefficient (gate G1)

```
theorem shift_leading_coeff {R : Type*} [CommRing R] [IsDomain R] (p : R[X]) :
    (p.comp (X + C 1)).coeff p.natDegree = p.leadingCoeff
```

Composition with $X + 1$ preserves the top coefficient; equivalently $[\theta^d]\, b(\theta+1) = \beta_d$. (The domain hypothesis is needed because the underlying Mathlib lemmas `natDegree_comp` / `leadingCoeff_comp` require `NoZeroDivisors`.)

T2 — edge polynomial closed form (gate G2)

```
theorem edge_poly_closed_form {R : Type*} [Field R] [CharZero R] (d : ℕ) (_hd : d ≠ 0)
    (βd c : R) :
    1 - βd * (-(c / (d : R))) ^ d
      = 1 + (-1) ^ (d + 1) * (βd / (d : R) ^ d) * c ^ d
```

With $a_{00} = 1$ and edge coefficient $a_{1,d} = -\beta_d$, the edge balance $a_{00} + a_{1,d}\,(z\,u^d) = 0$ gives

$$\chi_d(c) \;=\; 1 - \beta_d\big(-\tfrac{c}{d}\big)^d \;=\; 1 + (-1)^{d+1}\frac{\beta_d}{d^d}\,c^d,$$

proved d-free: the lemma is stated for general d with the single hypothesis $d \neq 0$ (equivalently $d \geq 1$ over $\mathbb{N}$). The manipulation $\big(-(c/d)\big)^d = (-1)^d c^d / d^d$ is valid for every $d \geq 1$ in characteristic 0.

T3a — affine-reindexing leading-coefficient law (gate G4)

```
theorem reindex_leading_coeff {R : Type*} [Field R] (p : R[X]) {d : ℕ}
    (hd : p.natDegree = d) (α γ: R) (hα : α ≠ 0) :
    (p.comp (C α* X + C γ)).coeff d = α^ d * p.leadingCoeff
```

The Lean conclusion is literally about the *degree-d coefficient* $[\cdot]_d$, written `.coeff d`, *not* `.leadingCoeff`. The two coincide here precisely because $\deg(b \circ \varphi) = d$ whenever $\alpha \neq 0$: the proof first establishes $\texttt{natDegree}(b \circ \varphi) = d$ (via `natDegree_comp` and $\deg(\alpha X + \gamma) = 1$), so the degree-$d$ coefficient *is* the leading coefficient of $b \circ \varphi$. Hence $\text{lead}(b \circ \varphi) = \alpha^d \beta_d$, *independent of* γ.

T3b — the ξ_0 covariance (gate G4)

```
noncomputable def xi0val (d : ℕ) (βd : ℝ) : ℝ:= (d : ℝ) / βd ^ ((1 : ℝ) / (d : ℝ))

theorem xi0_covariance (d : ℕ) (hd : d ≠ 0) (α βd : ℝ) (hα : 0 < α) (hβ : 0 < βd) :
    xi0val d (α ^ d * βd) = α⁻¹* xi0val d βd
```

With $\xi_0(\beta_d) = d/\beta_d^{1/d}$ (real `rpow`), the rescaling $\beta_d \mapsto \alpha^d \beta_d$ scales ξ_0 by α^{-1}. **Scope (corrected).** This is an identity about the *real positive value* ξ_0 as a function of the leading coefficient; it asserts *nothing* about the roots of χ_d. The root-modulus fact G3 — that every complex root c of χ_d has $|c| = \xi_0$ — is *VERIFIED* in computer algebra (§5) but is *not* formalized in Lean (§6). The numerical value of ξ_0 does equal that common root modulus, but `xi0_covariance` captures only the covariance of ξ_0, not the root-modulus statement.

Sign regime ($\beta_d > 0$). T3b carries the hypotheses $\alpha > 0$ and $\beta_d > 0$: real `rpow` is well-behaved (and the `Real.mul_rpow`/`Real.rpow_mul` step valid) only for nonnegative bases. The abstract's object-level hypothesis on a PCF is $\beta_d \neq 0$, which is strictly weaker. **The formalized covariance therefore covers $\beta_d > 0$ only**; the case $\beta_d < 0$ is outside the Lean scope. This is not a gap in the mathematics but in what is formalized: for $\beta_d < 0$ the relevant object is the common *complex* root modulus $|c| = d/|\beta_d|^{1/d}$ (real `rpow` of a negative base is defined in Mathlib but is not the analytic continuation one wants here), which is exactly the complex root-modulus fact G3 — VERIFIED in computer algebra (for $\beta_d < 0$ the modulus $|c| = d/|\beta_d|^{1/d}$ was re-checked here for $d = 2..7$, all d roots to 30 digits), not Lean-formalized. A referee asking "what happens for negative leading coefficient?" is answered: it is handled by the VERIFIED complex-modulus G3, deliberately not by the PROVEN real-`rpow` T3b. This $\beta_d > 0$ regime is *inherited by the conditional end statement*: `borelRadius_covariance` carries `hlc : 0 < p.leadingCoeff` (i.e. $\beta_d > 0$), which is exactly the hypothesis that both its second `edge_implies_borel_radius` application (on $b \circ \varphi$, whose leading coefficient $\alpha^d \beta_d$ must be positive) and its `xi0_covariance` call require. So for a reader who cares only about the end statement, $\beta_d > 0$ bites precisely there, through `hlc`; it is load-bearing, not decorative.

3 The $\mathsf{H}_{\mathsf{edge}}$ axiom and the conditional end statement

The analytic input is isolated as two explicit declarations, so that anything depending on it is flagged in its axiom cone:

```
-- ξ₀read off a real polynomial: the same real function as in T3b.
noncomputable def xi0 (p : ℝ[X]) : ℝ:= xi0val p.natDegree p.leadingCoeff

axiom borelRadius : ℝ[X] → ℝ-- OPAQUE; not constructed
```

```
axiom edge_implies_borel_radius (p : ℝ[X]) (hd : 2 ≤ p.natDegree)
    (hlc : 0 < p.leadingCoeff) : borelRadius p = xi0 p -- this is H_edge
```

Here `xi0` p is *defined* as `xi0val(` p`.natDegree, `p`.leadingCoeff )`, i.e. the same real function $\xi_0 = d/\beta_d^{1/d}$ from T3b, now evaluated at the degree and leading coefficient of p; this is the only place ξ_0 enters the polynomial-level statement. `borelRadius` is the analytic Borel-singularity radius, left *opaque* (we do not construct it). `edge_implies_borel_radius` is exactly $\mathsf{H}_{\mathsf{edge}}$: for a degree-$\geq 2$ denominator with positive leading coefficient, the analytic radius equals the algebraically computed ξ_0. It is *not* proved. The conditional theorem is then

```
theorem borelRadius_covariance (p : ℝ[X]) (α γ: ℝ) (hd : 2 ≤ p.natDegree)
    (hlc : 0 < p.leadingCoeff) (hα : 0 < α) :
    borelRadius (p.comp (C α* X + C γ)) = α⁻¹* borelRadius p
```

i.e. $\xi_0(b \circ \varphi) = \alpha^{-1}\xi_0(b)$ for $\varphi(n) = \alpha n + \gamma$, $\alpha > 0$ — **proven conditional on $\mathsf{H}_{\mathsf{edge}}$**. Its proof composes $\mathsf{H}_{\mathsf{edge}}$ (applied to b and to $b \circ \varphi$) with the purely algebraic T3a and T3b; no analytic content is smuggled in.

4 Soundness: axiom cones, build, toolchain

The decisive check, under the project's discipline, is the axiom cone, not a green build (a `sorry` compiles). The cones below were obtained from `#print axioms` by compiling against the pinned toolchain.

Declaration	Axiom cone	Status
`shift_leading_coeff`	{`propext`, `Classical.choice`, `Quot.sound`}	PROVEN
`edge_poly_closed_form`	{`propext`, `Classical.choice`, `Quot.sound`}	PROVEN
`reindex_leading_coeff`	{`propext`, `Classical.choice`, `Quot.sound`}	PROVEN
`xi0_covariance`	{`propext`, `Classical.choice`, `Quot.sound`}	PROVEN
`borelRadius_covariance`	{`propext`, `Classical.choice`, `Quot.sound`} ∪ {`borelRadius`, `edge_implies_borel_radius`}	COND.

The first four cones are *exactly* the clean set, with no `sorryAx`. The end statement's cone additionally exposes the two $\mathsf{H}_{\mathsf{edge}}$ axioms — expected and correct, and the precise sense in which the analytic claim remains undischarged.

Build / hygiene. `lake build` completes with exit code 0. The sources contain zero `sorry`, `sorryAx`, `admit`, or `native_decide` (the only textual occurrence of "sorryAx" is inside a documentation comment).

Toolchain (pinned, reproducible). Lean `leanprover/lean4:v4.29.0`; Mathlib rev `8a178386ffc0f5fef0b77738bb5449d50efeea95` (inputRev `v4.29.0`), committed via `lake-manifest.json`. The Mathlib lemmas relied upon (each `#check`-confirmed in the installed source) include `natDegree_comp`, `natDegree_X_add_C`, `leadingCoeff_comp`, `monic_X_add_C`, `natDegree_linear`, `leadingCoeff_linear`, `Real.rpow_natCast`, `Real.mul_rpow`, `Real.rpow_mul`, and `Real.rpow_one`; no lemma name was used from memory.

5 Numeric/symbolic ground truth (VERIFIED, gate G1–G4)

Before formalizing, the load-bearing identities were checked in computer algebra; all four gates were *re-run in this session* (sympy for the identities; `mpmath` for the G3 root moduli) for $d = 2..6$, with $d = 7$ spot-checks, so the Lean *statements* match a verified gate:

Gate	Statement	Checked
G1	$[\theta^d]\, b(\theta+1) = \beta_d$	$d = 2..6$
G2	$1 - \beta_d(-(c/d))^d = 1 + (-1)^{d+1}(\beta_d/d^d)c^d$	$d = 2..6$
G3	every root c of χ_d has $\lvert c\rvert = d/\beta_d^{1/d} = \xi_0$	$d = 2..6$
G4	$\mathrm{lead}(b\circ\varphi) = \alpha^d\beta_d$; $\xi_0(b\circ\varphi) = \alpha^{-1}\xi_0(b)$	$d = 2..6$

G3 at $d = 3$ in particular settles, numerically, the previously-deferred direct cubic check (re-confirmed here for $d = 2..6$, $d = 7$ spot-checked, by computing all d complex roots and verifying $|c| = \xi_0$ to 30 digits). **G3 is deliberately *not* reflected on the Lean side**: T3b is the covariance of the real value ξ_0, not a root-modulus statement (§6). All four gates are *VERIFIED*, a grade strictly weaker than the PROVEN Lean core and recorded as such.

6 What is *not* proved (scope and "could not confirm")

- **The analytic $\mathsf{H}_{\mathsf{edge}}$.** The edge $\Rightarrow$ Borel-radius implication (Wasow §19 sectorial existence; Birkhoff–Trjitzinsky Borel summability) is *cited-conditional*: encoded as an axiom, not formalized. No analysis/asymptotics content of Mathlib was used to discharge it. This is the single load-bearing analytic assumption.
- **Full complex root-set of χ_d (G3).** We do *not* formalize "every complex root of χ_d has modulus ξ_0" in Lean at all — it is *VERIFIED* only in computer algebra. T3b (`xi0_covariance`) formalizes the covariance of the *real value* ξ_0, which is a strictly different statement and must not be read as capturing G3. This is a deliberate scope choice, not a hidden claim.
- **No $d \geq 7$ or all-d analytic claim.** The VERIFIED gate is $d \leq 6$; the PROVEN T1–T3 are d-free as algebraic identities, but the analytic ξ_0 interpretation rests on $\mathsf{H}_{\mathsf{edge}}$ regardless of d.
- **Could-not-confirm list:** empty for Mathlib lemma names (every lemma used was located and `#check`ed in the installed source).

Honest one-line claim. *The finitary algebra of the M9 Newton edge — shift-invariance, the closed form of χ_d, the γ-free affine leading-coefficient law, and the α^{-1} covariance of $\xi_0 = d/\beta_d^{1/d}$ — is machine-checked PROVEN in Lean 4 with a clean axiom cone; that the analytic Borel radius equals ξ_0 and inherits the covariance is PROVEN only conditional on the explicit, unproved hypothesis H_{edge}.*

7 Reproducibility

The axiom-cone table in §4 is the load-bearing soundness claim, and — as a `sorry` compiles — it is established only by an actual `#print axioms` run, not by a green build. To reproduce it exactly:

1. Clone the repository and use the pinned `lean-toolchain` (`v4.29.0`) and `lake-manifest.json`.
2. `lake exe cache get` (prebuilt Mathlib oleans), then `lake build` (exit code 0).
3. Run the committed checker against the pinned toolchain: `lake env lean cone_check.lean`. It imports `PcfEdge` and prints `#print axioms` for all five declarations. The output below was **captured live** on 2026-06-05 by the run above, on a clean clone of the committed repository (decl names are emitted namespaced as `PcfEdge.<name>`; shown un-namespaced for width):

```
-- CAPTURED live on a clean clone (2026-06-05) via: lake env lean cone_check.lean
-- decl names emitted namespaced as PcfEdge.<name>; shown un-namespaced for width.
'shift_leading_coeff' depends on axioms: [propext, Classical.choice, Quot.sound]
'edge_poly_closed_form' depends on axioms: [propext, Classical.choice, Quot.sound]
'reindex_leading_coeff' depends on axioms: [propext, Classical.choice, Quot.sound]
'xi0_covariance' depends on axioms: [propext, Classical.choice, Quot.sound]
'borelRadius_covariance' depends on axioms:
  [propext, Classical.choice, borelRadius, edge_implies_borel_radius, Quot.sound]
```

   This is exactly the §4 table: the first four cones are the clean set, and only the conditional end statement exposes the two $\mathsf{H}_{\mathsf{edge}}$ axioms.

Provenance of the §4 cones (read this with the table). The §4 cones were **captured live** on 2026-06-05 by running the committed checker `cone_check.lean` on a *clean clone* of the committed repository against Mathlib rev `8a17838...` (`lake exe cache get`; then `lake build`, which completed all 8250 jobs at exit 0; then `lake env lean cone_check.lean`). The source `PcfEdge/Basic.lean` (SHA-256 `cac02d77...1c8dd48a7`) was re-verified unchanged against the bytes built. No `sorryAx` appears in any cone; the only non-standard axioms are `borelRadius` and `edge_implies_borel_radius` ($\mathsf{H}_{\mathsf{edge}}$), present by design in the conditional end statement only. This was the **required** pre-tag cone gate, and it has **passed**; the captured record, with the byte-level command transcript, is preserved in `AXIOM_CONES.txt`.

Relation to the deposited record. This artifact is a companion to the M9 preprint (concept DOI `10.5281/zenodo.20542161`), whose own statement notes that "there is no Lean formalization here" and that the general-k step is "a hand induction ($k = 0..8$ machine-checked in sympy)". The present development closes precisely that gap for the finitary core, while preserving the paper's cited-conditional treatment of the analytic edge $\Rightarrow$ radius step.

AI-assistance disclosure. Prepared with AI assistance (GitHub Copilot; Anthropic Claude) for Lean drafting, computer-algebra verification, and manuscript preparation. All axiom cones were obtained by compiling against the pinned toolchain; the author takes responsibility for the content and for the placement of each claim within the four-grade epistemic partition.